

Serzhong and Menchuna Non-wood Forest Product Management and Marketing Plan Phuentenchhu Gewog



Tsirang Dzongkhag
Plan period: August 2021- July 2026

PART A: NWFP management and marketing plan

1. Introduction

The Social Forestry and Extension Section (SFES) in the Division, carried out awareness on NWFP management for communities living in Serzhong chiwog under Puentenchu gewog in May 2021. Subsequently with overwhelming interest from the communities, SFES spear headed with NWFP management planning and marketing exercise during Fiscal year 2020-2021. For this, Social Forestry and Extension Division provided financial support. Thus, Serzhong and Menchuna NWFP Management group was formed. The group will manage 731.79 ha (1807.52) acre of NWFP resources in State Reserved Forest Land, benefitting 20 households.

2. Goals & Objectives

Goals:

- ✓ Management and utilization of NWFP resources enhancing income and employment generation for the community.

Objectives:

- ✓ To manage and utilize NWFPs sustainably
- ✓ To create employment opportunity
- ✓ To generate income
- ✓ To domesticate and propagate potential NWFPs

3. Description of NWFPs

There are many Non-Wood Forest Products (NWFPs) in Serzhong chiwog under Puentenchu gewog. The NWFPs ranges from medicinal plants, cane, bamboo, mushrooms, dyes, gums, fruits, essential oils, fibers, ferns and many others. In the absence of adequate technical capacity within the communities on NWFPs, particularly knowledge on resources and their ecology and availability, habitats, opportunities and threats, the rural people has been not following the principle of protection, sustainable utilization and management. Generally, among the resources optimally used for income generation from forest are Timber and NWFPs. The NWFPs play an important role in the daily lives and overall wellbeing of Bhutanese people especially in rural areas. Canes (*Plectocomia himalayana*) has been one of the most widely cultivated cash crop in Menchuna village. People use cane shoot as a delicacy in vegetable preparation and has high market value. Serzhong chiwog has the potential to propagate canes in the wild and protect their natural regeneration. They can be cultivated on a commercial scale to generate cash income for the group; few farmers have planted on small scale, mostly at household level at the edge of their fields. Varieties of bamboo species grow naturally in the NWFP management area and has good potential to generate income in future.

Table 1: Brief description of NWFPs for commercial harvesting

<i>NWFP name</i>	<i>Harvesting area (s) (name)</i>	<i>Parts harvested</i>	<i>Use purpose (traditional & others)</i>	<i>Harvesting season (months)</i>	<i>Estimated quantity that can be collected</i>
<i>Diplazium esculentum</i> (Ningro)	-	Young shoots	Vegetable	March-September	
(Simrai saag)	-	Stem/leaves	Vegetable/Medicine	All year	
<i>Piper hamiltonii</i> (Jungale Pan)	-	Leaves	Stimulant	All year	
<i>Phyllanthus emblica</i> (Amala)	-	Fruits	Pickle	November - February	
<i>Dioscorea spp</i> (Jungale torul)	-	Tubers	Food/offerings		
<i>Plectocomia himalayana</i> (Phekray)	-	Shoot	Vegetable/Medicine/handicraft	Winter	
<i>Bambusa nutans</i> (Mal bans)	-	Shoot/stem	Pickle/handicraft	Depends on species type. Refer table 2	

4. Harvesting prescriptions for sustainable NWFP harvesting

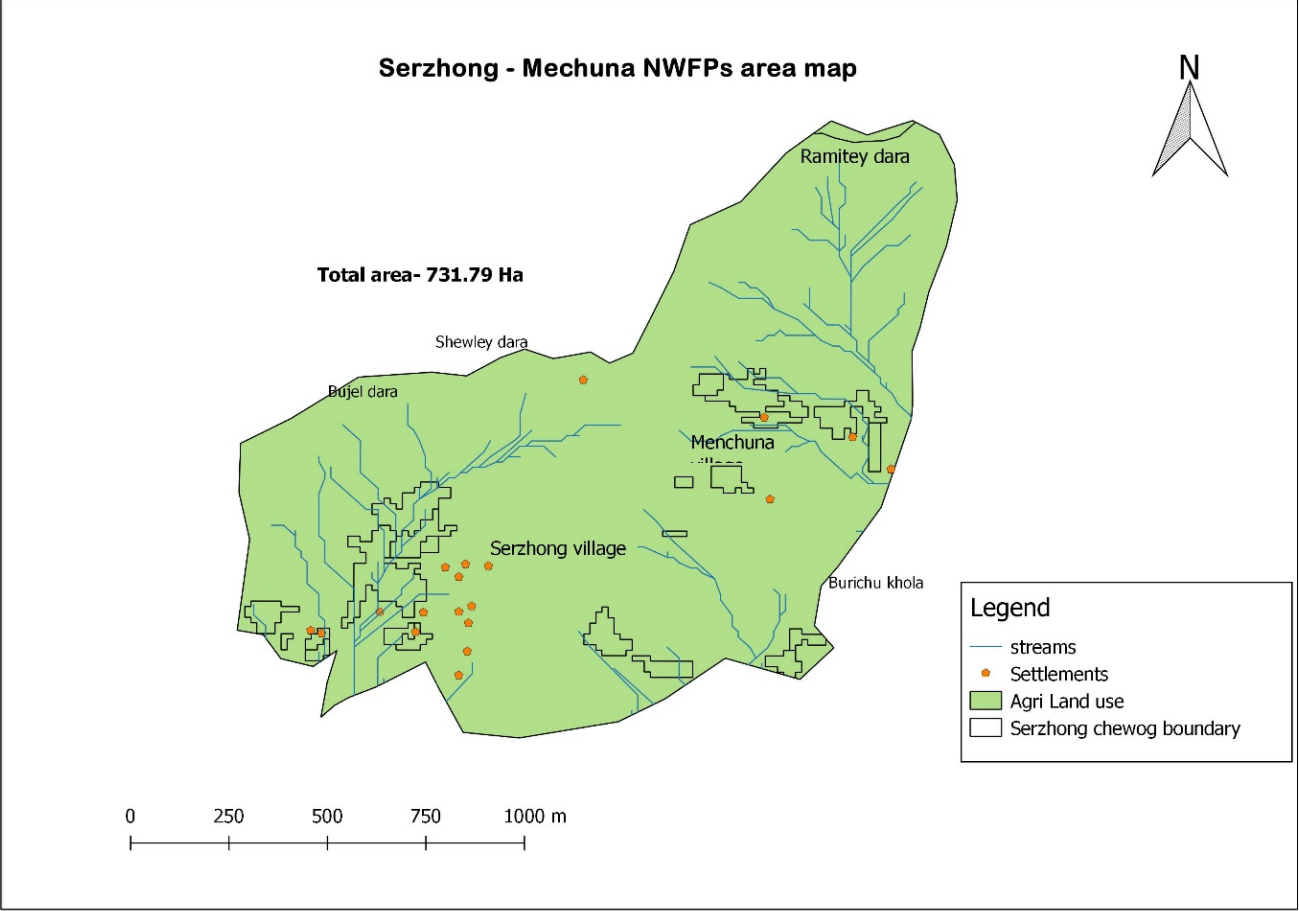
Table 2: Harvesting prescriptions for sustainable harvesting

<i>NWFP Species</i>	<i>Harvesting prescriptions for sustainable harvesting</i>
1. <i>Diplazium esculentum</i> (Ningro)	Harvesting time- March to September • Harvests young shoot that are 15cm length. • Don't harvest the entire shoot coming up in season. • Retain at least 50% of young shoot of last flush for the development of leaves. • Don't harvest the shoots that have developed to leaves. • Don't cut down trees which provide shade to the plant.

2. <i>Plectocomia himalayana</i> (Phekray)	Harvesting time- Winter time for mature stems For mature stems- • Cut cane stem only when mature, to be recognized by: • Loosening of leaf sheath from the lower part of the cane stem • Exposure of lowest part of the stem • Turning to yellow-reddish along the entire stem length • Diameter over the entire length of the cane is the same (the diameter at the lower part of the cane is reduced) • Pull down as much of the stem as possible (avoid leaving cane parts high up in the tree) • Cut only canes with a usable length of at least 10 m. Discard at maximum 1.2 to 1.5 meters from the top end. • Harvest during winter as canes have a better quality during this season (lower water content and more durable) For cane shoots- • Cut young shoot by bending the cane stem with the help of a long and strong stick with a hook • Do not cut the entire cane to collect the shoots Cut the leading shoot with a sharp knife diagonally (not straight in order to avoid rotting of the remaining stump); leave lateral shoots • Care shall be taken to avoid damaging sucker buds (these will later develop into lateral shoots) For both mature stems and cane shoots- • Uprooting, cutting of rhizomes / suckers shall be avoided. • Trees supporting the canes shall not be felled • The debris after harvesting canes should be taken outside the cane area in order to avoid infection with insect borers or fungi.
3. <i>Piper hamiltonii</i> (Jungal pan)	Harvesting time- Throughout year • Collect leaves only by hand picking without damaging the climber and its tender buds. • Harvesting of leaves from a single (same) climber shall be allowed to maximum of 4 times per year. • Collect at maximum 80% of the leaves from a single climber. • Do not cut down the host tree to ease harvesting.
4. <i>Phyllanthus emblica</i> (Amala)	Harvesting time- November to February • Harvest fruits by shaking the tree, handpicking and / or use bamboo poles with a hook. • Do not cut down trees for the harvesting of fruits

5. <i>Dendrocalamus hamiltonii</i> (Choya bans)	Harvesting time- For culms (November to April) For culms – • Cut culms above one node from the ground level • Do not cut culms less than one-year-old • Retain 50% of healthy culms from a clump • Harvest culms in condensed / congested clump in “U” or horseshoe form. Use otherwise single culm selection method For shoots - • Collect bamboo shoots to a maximum of 25% from a clump • Cut bamboo-shoots with a sharp knife at the ground level without damaging or wounding the other healthy ones • After cutting and collecting the bamboo-shoot, cover that void place with top soil • Protect the remaining shoots and let them further develop into culms
6. <i>Bambusa clavata</i> (Chiley bans)	Harvesting time- For culms October to April • Cut culms above one node from the ground level • Do not cut culms less than one-year-old • Retain 50% of healthy culms from a clump
7. <i>Bambusa nutans</i> (Mal bans)	Harvesting time - For shoots – May to September • Harvest culms in condensed / congested clump in “U” or horseshoe form. Use otherwise single culm selection method For shoots: • Collect bamboo shoots to a maximum of 25% from a clump • Cut bamboo-shoots with a sharp knife at the ground level without damaging or wounding the other healthy ones • After cutting and collecting the bamboo-shoot, cover that void place with top soil • Protect the remaining shoots and let them further develop into culms
8. <i>Dioscorea spp</i> (Jungal torul)	Harvesting time- Jan to April • Dig at the root of the plant. Collect all yam. • Leave the fruits to fall on the ground and germinate into new plants
9. <i>Thysanolaena latifolia</i> (Amlisho)	Harvesting time- December to March • Cut the foliage, inflorescence at 15 cm from the ground • Do not damage the rhizomes in order to allow natural regeneration
10. <i>elatostema lineolatum</i> (Chotey gaglata/Damroo)	Harvesting time- Any time • Collect plants that are 15cm and above • Retain stump of around 3” for sprouting of new shoots • Don’t uproot whole plant • Don’t cut down trees and create more openings • Don’t disturb its catchment areas

Serzhong and Mechuna NWFP harvesting area map



5. Marketing and sales plan

The NWFP products such as fern top, bamboo, wild yam and cane shoot harvested by the group can be sold to retailers in Damphu town and fetch income to benefit the community. The estimated quantity, price per unit and the total value of expected sales are reflected in table 3.

Table 3: Marketing and sales plan

NWFP	Potential market	Estimated quantity for sale per season (unit)	Expected price per unit	Total (Nu)
Fern	Damphu town	500 bundles	20/ piece	10,000.00
Bamboo	Geog/Damphu	350 pieces	80/ piece	28,000.00
Wild yam	Damphu	200kg	100/ kg	20,000.00
Cane shoot	Damphu	250 piece	50/ piece	12,500.00

6. Marketing arrangements

Harvesting and marketing of NWFPs in a season will be done as a group and 30 % of the income generated through sale of the products shall be deposited into groups account and rest will be divided equally among the NWFP Management group members. The daily allowance for the sales -man and the transportation cost shall be borne from the sale of the products. If the products could not be sold at the local markets, NWFP Management committee members will take the products to other Dzongkhags for marketing.

7. Benefits to the members

Out of several NWFPs proposed for sale, only four products are selected based on their marketing potentiality. The average income calculated for individual members is found with Nu.2,467.50/head as shown in table 4. Nevertheless, the group as a whole is contributing 30% of their total sale proceeds as a group fund i.e. Nu.21,150/occasion as saving and the rest 70 % will be divided among the members. This will directly benefit the group to generate cash income.

Table 4: Financial Projections of benefits to the members from NWFP sales

Sl. no	Particulars	Products name				Total amount (Nu)
1	Revenues	Fern	Cane shoot	Wild yam	bamboo	
a	Total quantity (unit)	500 bundles	250 pes	200 kg	350 pes	1300.00
b	Market price(Nu per unit)	20	50	100	80	250.00
c	Total revenues from forest product sale (a*b)	250 x 20	150 x 50	100 x 100	200x80	70,500.00
A	Total revenue	10,000.00	12,500.00	20,000.00	28,000.00	70,500.00
2	Expenses					
a	Packing materials	-	-	-		
b	Royalty					
c	Transportation cost	5000	-	-		5000.00
d	Compensation fee to executive committee for marketing products	2000	-	-		2000.00
e	Contribution to community fund	21,150.00	-	-		21,150.00
B	Total expense	28,150.00				28,150.00
C	Total benefit (net income) of group members (Total revenue A –Total expense B)	49,350.00				49,350.00
D	Average income to group members (C/ no of group members	2,467.50	-	-		2,467.50
E	Price per unit paid to group members (C/1a)	37.96	-	-		37.96

8. Benefits to the group

The Table 5 below shows Financial projection of the benefits to the group as a means for the community development fund. The total revenue is collected from annual membership fees and from the sale proceeds of products sold in the market such as fern, cane, wild yam and bamboo by the NWFP group. The expenses included transportation cost and compensation fee to the executive members for marketing the products. It is assumed that, a sum of Nu.18,150.00 will remain as fund in the group account on annual basis. This fund can be used for NWFP developmental activities as per their specific annual work plan and benefit the group in particular.

Table 5: Financial projection of the benefits to the group, the community development fund

Sl, no	Particulars	Amount (Nu)
	Revenues	
1	Membership fees	4000.00
2	Contribution to the Community Development Fund (CDF) from sale of produce 1	3000.00
3	Contribution to the CDF from sales of product 2	3750.00
4	Contribution to the CDF from sales of product 3	6000.00
5	Contribution to the CDF from sales of product 4	8400.00
A	Total revenues	25,150.00
	Expenses (only include expenses that will be paid from the group funds)	
1	Transportation cost	5000.00
2	Compensation fee to executive committee for marketing products	2000.00
B	Total Expenses	7000.00
C	Balance: Total revenues – Total expenses	18,150.00

At present, the financial benefit reflected as Nu.18,150.00 is quite low for the group. However, it is projected to generate gradual increase of the fund as more bamboo, cane and broom grass will be planted and managed from 2022 as per the work plan.

Work plan for the Group:

Activities	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Broom grass plantation						x	x					
Cane plantation						x	x					
Nursery development for Cane										x		
Bamboo plantation						x						
Marketing of potential NWFPs						x						x

PART B: By-laws: Group regulations and norms

1. Members of the NWFP management & marketing group	Village	Number of Households	Geog
	Serzong and Menchuna	20 HHs	Phuentenchhu
	Responsibilities and Rights of members: ✓ Management of the NWFPs ✓ Monitoring of NWFP development activities ✓ Attend meeting/contribute labor force when required ✓ Control cattle from entrance inside NWFP area ✓ Permit to harvest and sale Broom as per the agreement and harvesting rules		

2. Executive committee			
2.1 Chairperson	Name of the Chairperson: Mr. Dhan Bdr. Rai (17556939)		
	Responsibilities of coordinator/Chairperson: ➤ Chair the meeting/decision making with members ➤ To resolve disputes ➤ Official correspondence ➤ Arrangement for visitors ➤ To inform members on the date and venue for carrying out the activities regarding NWFP group ➤ To coordinate the sale of NWFP product ➤ Preparation of annual and final report (5 years)		

2.2 Secretary	Name of the Secretary: Mr. Guman Singh Subba(17674688)		
	Responsibilities of secretary: ❖ Officiate in absence of coordinator ❖ Supervise work plan activities ❖ Meetings minutes keeping ❖ Record keeping ❖ Assist treasure during fund transaction		

2.3 Treasurer	Name of the Trasurer: Mr. Dasarath sanyasi(17317223)		
	Responsibilities of Treasurer:  Financial transaction with Coordinator  Maintain cash transaction records  To collect membership fees and penalties  Financial transections		

3. Membership fee and financial arrangements	-Members will contribute the entry fee of Nu.200/- annually -Financial activities will be met from penalties and entry fees. -Collected penalties and fees will be deposited in group’s account maintained with BDBL
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4. Conflict Resolution	-Chairperson, Secretary and Treasurer will try to settle the case -Forward the case to the Gup, Geog Administration -CFO -District Court
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5. Penalties	➤ If the members fails to attend meeting or labor contribution, he/she will levy fine of Nu. 500/- per occasion ➤ Fine will be levy to those owners who let cattle inside NWFP area as follows: <table border="1"> <tr> <th>Members</th><th>Outsiders</th></tr> <tr> <td>Fine of Nu.400/cattle with statement</td><td>Fine of Nu.500/cattle with written statement</td></tr> <tr> <td>Note: Members are allowed to collect and consume the NWFPs freely for domestic consumption after obtaining collection permit</td><td>If outsider does illegal work inside NWFP area he/she will pay the fine as described below.</td></tr> </table>	Members	Outsiders	Fine of Nu.400/cattle with statement	Fine of Nu.500/cattle with written statement	Note: Members are allowed to collect and consume the NWFPs freely for domestic consumption after obtaining collection permit	If outsider does illegal work inside NWFP area he/she will pay the fine as described below.
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Fine of Nu.400/cattle with statement	Fine of Nu.500/cattle with written statement						
Note: Members are allowed to collect and consume the NWFPs freely for domestic consumption after obtaining collection permit	If outsider does illegal work inside NWFP area he/she will pay the fine as described below.						

6. Fines and penalties

Sl.no	Products	Members	Non-members	Remarks
1	Bamboo	100/culm	500/culm	NWFPs are free for domestic consumption by members, however, they should obtain permit before collection. For commercial sale, it has to be done by group.
2	Cane	150/piece	250/piece	
3	Broom grass	50/bundle	250/bundle	
4	Timbur	50/kg	250/kg	
5	Fern	20/bundle	40/bundle	
6	Pan	20/bundle	50/bundle	
7	Simrai saag	25/bundle	50/bundle	

a. Membership arrangement

- All farmers of Serzhong chiwog shall be allowed to join as member in the NWFP Management group.
- In case of households who couldn't join as member during the formation of group shall pay the sum equivalent to amount deposited by a member till date plus Nu.3000.00. However, in case of separated house hold from family (chut-bhai), if all brothers or chut-bhais are living in the same chiwog since the formation of NWFP group. he/she can become a member after paying the sum equivalent to amount deposited by a member till date.
- For those people, who purchase the land at Serzong chiwog and want to settle there, after the formation of group, he/she can join the group after paying the sum equivalent to amount deposited by a member till date plus Nu.3000.00

b. Benefit sharing

- Broom grass: The product will be sold in the market and the sale proceed shall be divide equally among members after depositing 50% in the group account.
- Fodder: The fodder will be shared and utilized equally among the members; however, if there is excess, they can open auction it to the customers.
- Bamboo: The product will be sold in the market and the sale proceed shall be divide equally among members after depositing 50% in the group account.

c. Leaving and re-entry in the group

- The NWFP Management group members engaged in any activity, in contravention of the provisions of the act, rules, by-laws or the management plan, shall leave the group without any benefit.
- If the member had to leave the group with genuine reason(s) he/she will receive money that is equivalent to the saving bank amount divided between the existing numbers of members on the day of resignation.
- Re-entry of the members who had left the group with genuine reason(s) shall be allowed to join the membership after repaying the amount that was taken from the group as incentive. He/she will be treated as new members and shall follow the bylaws. However, those who left without reason shall not be allowed to join in any case. Applications should be submitted to the chairperson, while joining as a new member and leaving the group.

d. Financial management

- Fund will be utilized during any work that arises related to NWFP. (Transportation, meetings, visitors, etc.)
- The group will not avail loan to members if the total amount does not exceed Nu. 20,000.00 (twenty thousand) in the account and the reserve amount to be kept at bank will be Nu.5,000.00 (five thousand).
- The group will contribute Nu. 2,000.00 (two thousand) only as condolence to any members if there is demise of anyone within the members.
- If guest or the visitors' grant donation to the group, the amount will be deposited to the group account.

e. Record keeping

The following records will be kept for monitoring the NWFP marketing activities and for proper documentation of NWFP books of accounts and other records.

1. Annual Work Plan and Progress Report
2. Seedling and Plantation Record
3. Attendance Register
4. Record of Penalty and Compensation
5. Collection Permit
6. Records of NWFP Utilization
7. Money Receipt
8. Cash Book
9. Meeting Minutes
10. Loan Record
11. Stock Register

f. Terms for Executive members

- The tenure for executive members shall be for five years upon which new committees will be elected.
- If any executive members do not abide by the by laws, he/she will be terminated from the executive members upon repaying of any amount misused.

g. Meeting

- The General meetings will be conducted twice in a year i.e. June and December.

h. Miscellaneous

- If there is any Geog development plan which is going to affect the area of NWFP, the concern authority will pay the compensation depending upon the damages cause to the NWFP.
- The NWFP management group shall issue no objection clearance to the geog for developmental activities after receiving compensation.

i. Movement and export of NWFPs

- The NWFP groups/ traders/ individuals transporting NWFPs from GRF, shall obtain a transit permit as per FNCRR (Appendix 3.3). The transit permits shall be issued by the nearest forest office.
- For the export of NWFP produce, the exporter shall obtain a movement order for export (Appendix 3.4). The export permits shall be issued by the nearest forest office.
- Groups/ traders/ exporters shall obtain a clearance from BAFRA as per rules.

j. Monitoring & Evaluation

- The CFO office shall be responsible for general monitoring of the available NWFP resources and the implementation of the framework in the field in collaboration with NWFP group on a regular basis (see appendix 4)
- RO/Gewog Ranger shall be responsible for the monitoring of the implementation of the NWFP Management & Marketing Plan and make records of the actual collected and sold quantities in collaboration with NWFP groups (see appendix 4)
- CFO shall be responsible for monitoring the movements of NWFPs
- CFO/RO shall prepare an annual monitoring report describing resource availability and marketed NWFPs in collaboration with NWFP group
- CFO/RO shall be responsible to conduct and coordinate a mid-term review and a final evaluation of the implementation of the NWFP management & marketing plan (see appendix 4)

7. Powers of the Department

The Department has the right to terminate the NWFP management and marketing plan in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.